

- 1 (a) Use the list of formulae (MF19) to find $\sum_{r=1}^n r(r+2)$ in terms of n , simplifying your answer. [2]

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- (b) Express $\frac{1}{r(r+2)}$ in partial fractions and hence find $\sum_{r=1}^n \frac{1}{r(r+2)}$ in terms of n . [4]

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- (c) Deduce the value of $\sum_{r=1}^{\infty} \frac{1}{r(r+2)}$. [1]